ENGINEERING ECONOMY

LECTURE 3

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F/P and *P/F* Factors: Notation and Equations

Factor		Find/Given	Standard Notation Equation	Equation with Factor Formula	Excel Function
Notation	**Name**				
$(F/P,i,n)$	Single-payment compound amount	F/P	$F = P(F/P,i,n)$	$F = P(1+i)^n$	$= FV(i\%,n,,P)$
$(P/F,i,n)$	Single-payment present worth	P/F	$P = F(P/F,i,n)$	$P = F(1+i)^{-n}$	$= PV(i\%,n,,F)$

P/*A* and *A*/*P* Factors: Notation and Equations

Factor Notation	Factor Name	Find/Given	Factor Formula	Standard Notation Equation	Excel Function
$(P/A,i,n)$	Uniform series present worth	P/A	$\frac{(1+i)^n-1}{i(1+i)^n}$	$P = A(P/A,i,n)$	$= \text{PV}(i\%,n,A)$
$(A/P,i,n)$	Capital recovery	A/P	$\frac{i(1+i)^n}{(1+i)^n-1}$	$A = P(A/P,i,n)$	$= \text{PMT}(i\%,n,P)$

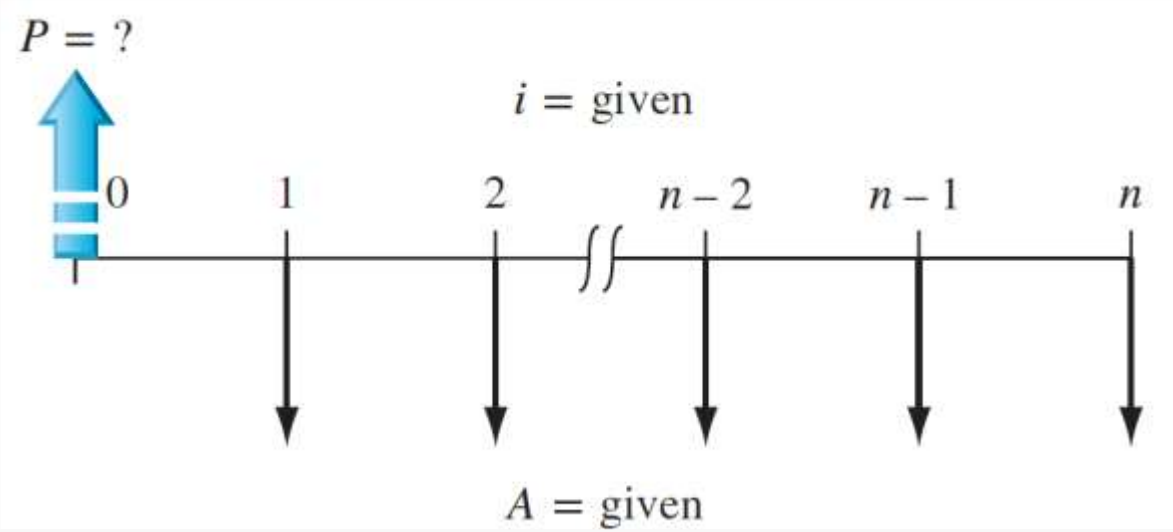
F/*A* and *A*/*F* Factors: Notation and Equations

Factor Notation	Factor Name	Find/Given	Factor Formula	Standard Notation Equation	Excel Functions
$(F/A,i,n)$	Uniform series compound amount	F/A	$\frac{(1+i)^n-1}{i}$	$F = A(F/A,i,n)$	$= \text{FV}(i\%,n,A)$
$(A/F,i,n)$	Sinking fund	A/F	$\frac{i}{(1+i)^n-1}$	$A = F(A/F,i,n)$	$= \text{PMT}(i\%,n,F)$

EXAMPLE

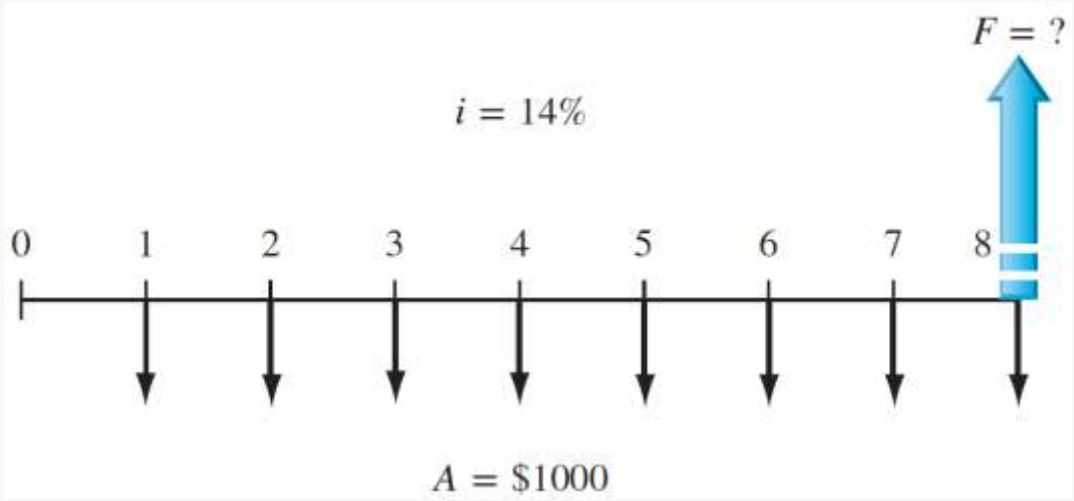
- How much money should you be willing to pay now for a guaranteed $600 per year for 9 years starting next year, at a rate of return of 16% per year?
- **Solution**
- The cash flows follow the pattern of the Figure, with A = \$600, i = 16%, and n = 9. The present worth is

$$P = 600(P/A,16\%,9) = 600(4.6065) = \$2763.90$$



EXAMPLE

- The president of Ford Motor Company wants to know the equivalent future worth of a $1 million capital investment each year for 8 years, starting 1 year from now. Ford capital earns at a rate of 14% per year.
- **Solution**
- The cash flow diagram shows the annual investments starting at the end of year 1 and ending in the year the future worth is desired. In $1000 units, the *F* value in year 8 is found by using the *F/A* factor.
- $F = 1000(F/A,14\%,8) = 1000(13.2328) = \$13,232.80$



EXAMPLE

- Suppose HBNA projected that $200 million investment can generate $50 million per year in revenue for 5 years starting 1 year after start-up. If 10% per year time value of money is given:
- (*a*) What is the equivalent future worth of the estimated revenues after 5 years at 10% per year?
- (*b*) Assume that, due to the economic downturn, the president predicts that the Corporation will earn only 4.5% per year on its money, not the previously anticipated 10% per year. What is the required amount of the annual revenue series over the 5-year period to be economically equivalent to the amount calculated in (*a*)?

- **Solution**

(*a*) Find *F*/*A* factor. P = \$200 million, A = \$50 million, n = 5, i = 0.1

$$F = 50(F/A,10\%,5) = 50(6.1051)$$

$$= \$305.255$$

(*b*) The *A*/*F* factor will determine the required A for 5 years. i = 0.045

$$A = 305.255(A/F,4.5\%,5) = 305.255\left[\frac{0.045}{(1.045)^5 - 1}\right] = 305.255(0.18279)$$

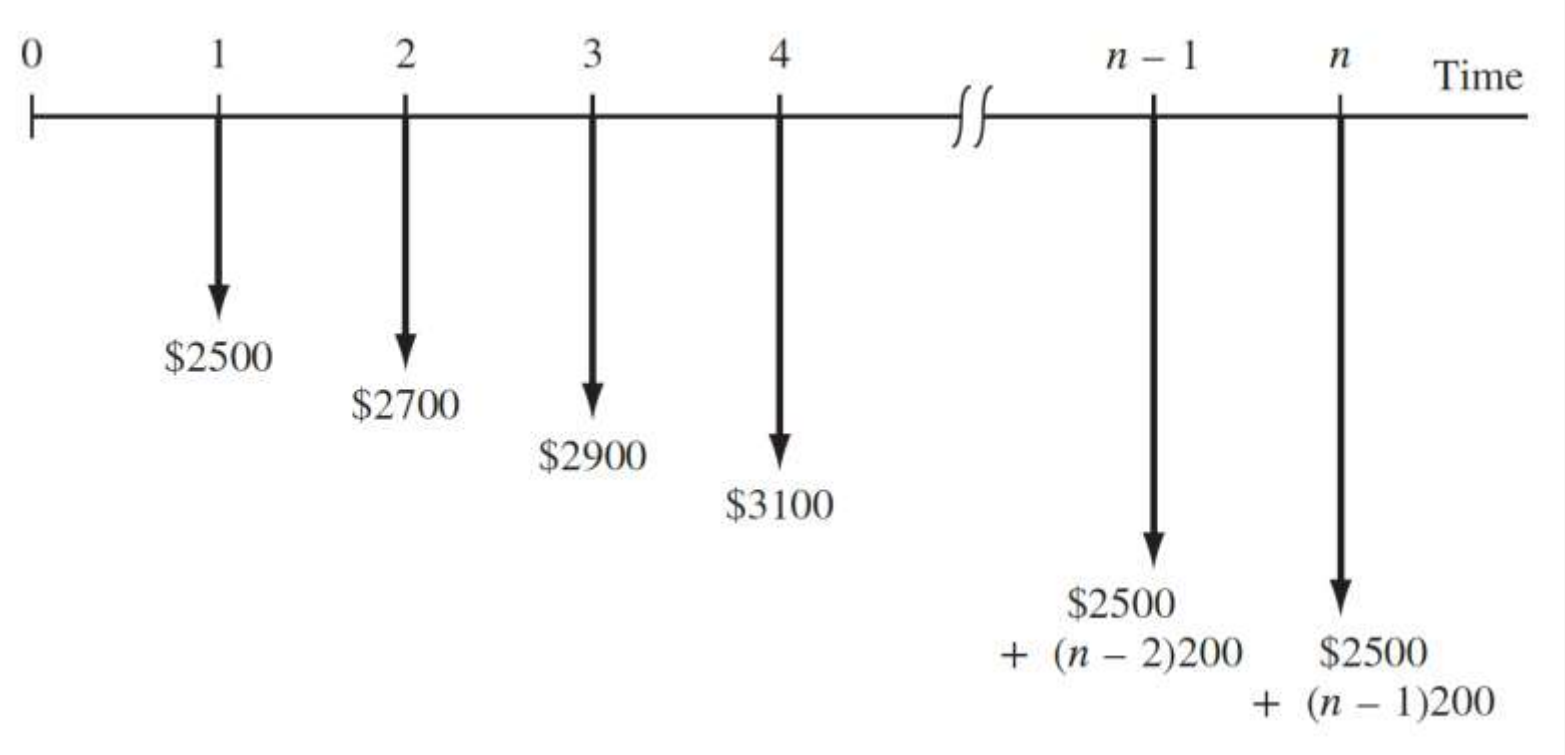
$$= \$55.798$$

Arithmetic Gradient Factors (P/G and A/G)

- An arithmetic gradient is a cash flow that either increases or decreases by a constant amount.
- The cash flow, changes by the same amount each period.
- The amount of decrease or the increase is the gradient or *G*.
- Therefore, it is composed of base amount and gradient part.

- Formulas previously developed for an *A* series have year-end amounts of equal value.
- In the case of a gradient, each year-end cash flow is different, so new formulas must be derived.
- First, assume that the cash flow at the end of year 1 is the **base amount** of the cash flow series and, therefore, not part of the gradient series.
- This is convenient because in actual applications, the base amount is usually significantly different in size compared to the gradient.
- For example, if you purchase a used car with a 1-year warranty, you might expect to pay the gasoline and insurance costs during the first year of operation.
- Assume these cost $2500; that is, $2500 is the base amount.
- After the first year, you absorb the cost of repairs, which can be expected to increase each year.

- If you estimate that total costs will increase by $200 each year, the amount the second year is $2700, the third $2900, and so on to year *n*, when the total cost is 2500 + (*n* − 1)200.
- The cash flow diagram is shown in the Figure.
- Note that the gradient ($200) is first observed between year 1 and year 2, and the base amount ($2500 in year 1) is not equal to the gradient.

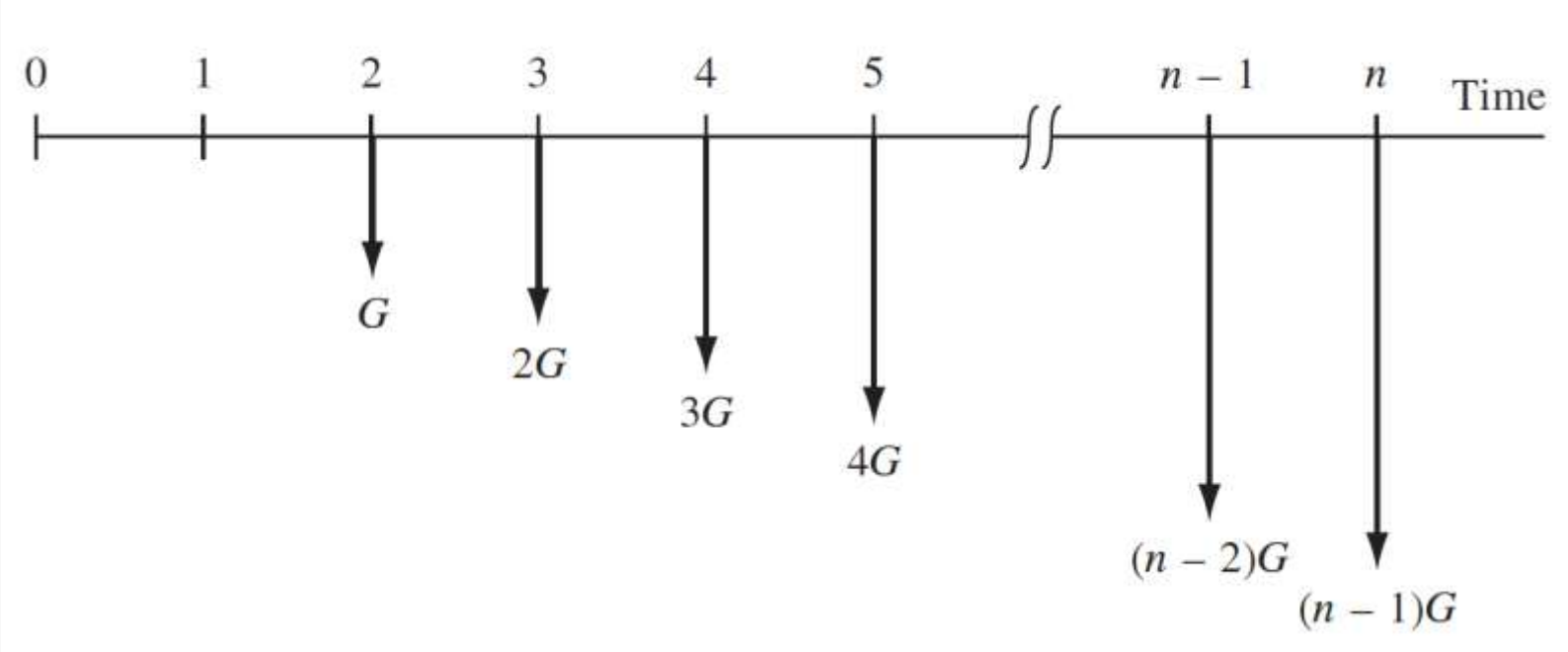


- Define the symbols *G* for gradient and *CFn* for cash flow in year *n* as follows:
- *G* = constant arithmetic change in cash flows from one time period to the next; *G* may be positive or negative.

$$\mathrm{CF}_n = \text{base amount} + (n-1)G$$

- It is important to realize that the base amount defines a uniform cash flow series of the size *A* that occurs each time period.
- We will use this fact when calculating equivalent amounts that involve arithmetic gradients.

- If the base amount is ignored, a generalized arithmetic (increasing) gradient cash flow diagram is as shown as:



- Note that the gradient begins between years 1 and 2. This is called a **conventional gradient.**

EXAMPLE

- A local university has initiated a logo-licensing program with the clothier Holister, Inc. Estimated fees (revenues) are $80,000 for the first year with uniform increases to a total of $200,000 by the end of year 9.
- Determine the gradient and construct a cash flow diagram that identifies the base amount and the gradient series.

The year 1 base amount is $CF_1 = \$80{,}000$, and the total increase over 9 years is

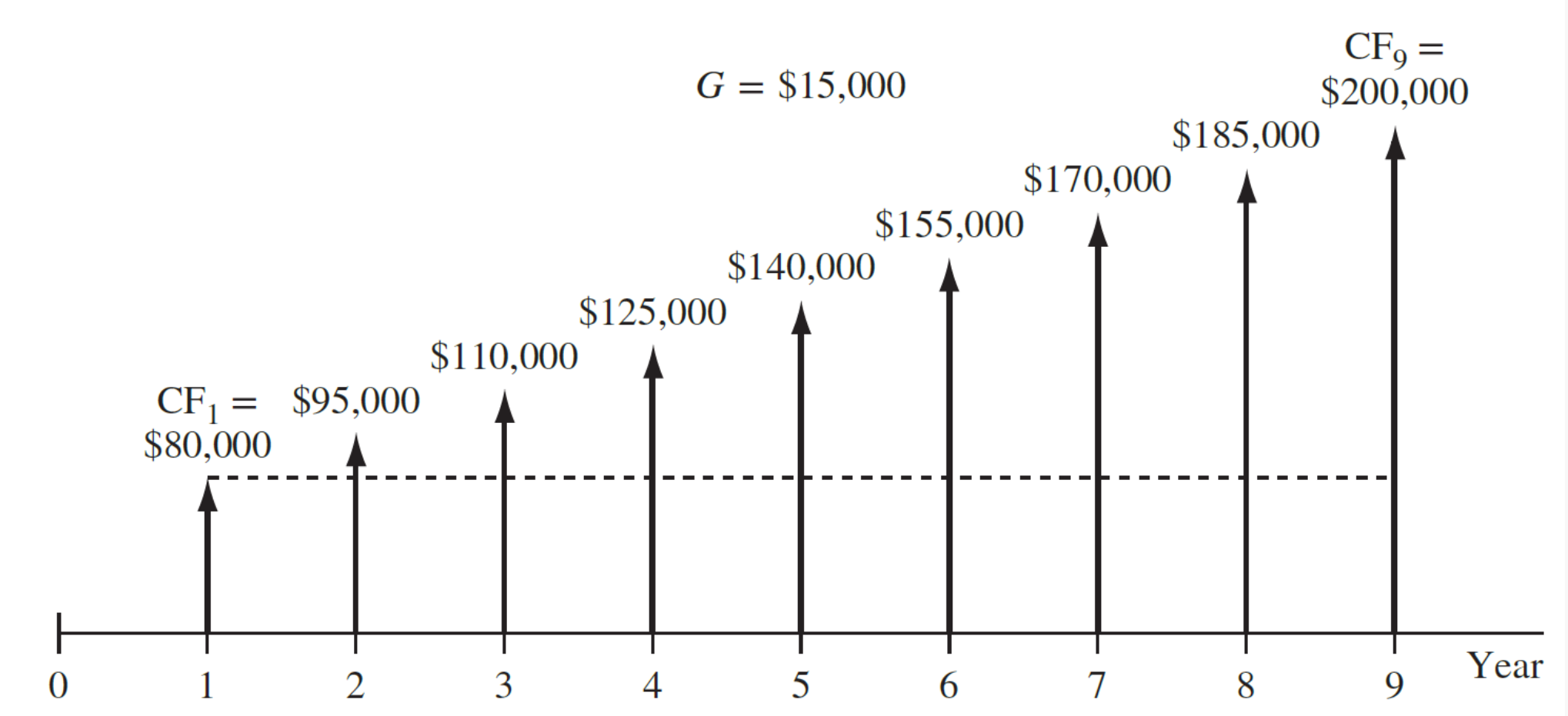
$$CF_9 - CF_1 = 200{,}000 - 80{,}000 = \$120{,}000$$

Equation $\mathbf{CF_n = \text{base amount} + (n-1)G}$

solved for G, determines the arithmetic gradient.

$$G = \frac{(CF_9 - CF_1)}{n-1} = \frac{120{,}000}{9-1}$$

$$= \$15{,}000 \text{ per year}$$



- The cash flow diagram shows the base amount of $80,000 in years 1 through 9 and the $15,000 gradient starting in year 2 and continuing through year 9.

- The **total present worth** *PT* for a series that includes a base amount *A* and conventional arithmetic gradient must consider the present worth of both the uniform series defined by *A* and the arithmetic gradient series. The addition of the two results in P_T .

$$P_T = P_A \pm P_G$$

- where P_A is the present worth of the uniform series only, P_G is the present worth of the gradient series only, and the + or − sign is used for an increasing (+*G*) or decreasing (−*G*) gradient, respectively.
- The corresponding equivalent annual worth A_T is the sum of the base amount series annual worth A_A and gradient series annual worth A_G, that is,

$$A_T = A_A \pm A_G$$

- Three factors are derived for arithmetic gradients: the *P*/*G* factor for present worth, the *A*/*G* factor for annual series, and the *F*/*G* factor for future worth. There are several ways to derive them. We use the single-payment present worth factor (*P*/*F*,*i*,*n*), but the same result can be obtained by using the *F*/*P*, *F*/*A*, or *P*/*A* factor.

$$P = G(P/F,i,2) + 2G(P/F,i,3) + 3G(P/F,i,4) + \cdots$$
$$+ [(n-2)G](P/F,i,n-1) + [(n-1)G](P/F,i,n)$$

$$\begin{aligned} P &= G\left[\frac{1}{(1+i)^2}\right] + 2G\left[\frac{1}{(1+i)^3}\right] + \cdots + (n-2)G\left[\frac{1}{(1+i)^{n-1}}\right] \\ &+ (n-1)G\left[\frac{1}{(1+i)^n}\right] \\ &= G\left[\frac{1}{(1+i)^2} + \frac{2}{(1+i)^3} + \cdots + \frac{n-2}{(1+i)^{n-1}} + \frac{n-1}{(1+i)^n}\right] \end{aligned}$$

$$P(1+i) = G\left[\frac{1}{(1+i)} + \frac{2}{(1+i)^2} + \cdots + \frac{n-2}{(1+i)^{n-2}} + \frac{n-1}{(1+i)^{n-1}}\right]$$

$$Pi = G\left[\frac{1}{(1+i)} + \frac{1}{(1+i)^2} + \cdots + \frac{1}{(1+i)^{n-2}} + \frac{1}{(1+i)^{n-1}}\right]$$

$$- G\frac{n-1}{(1+i)^n}$$

$$Pi = G\left[\frac{1}{(1+i)} + \frac{1}{(1+i)^2} + \cdots + \frac{1}{(1+i)^{n-2}} + \frac{1}{(1+i)^{n-1}} + \frac{1}{(1+i)^n}\right]$$

$$- G\frac{n}{(1+i)^n}$$

$$P = \frac{G}{i}\left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n}\right]$$

$$P = \frac{G}{i}\left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n}\right]$$

$$\frac{1}{i}\left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n}\right]$$

- is the arithmetic-gradient present worth factor or *P/G* factor that converts the arithmetic gradient (without base amount) for n years to the present worth at year 0. The standard notation is (*P/G*, *i*%, *n*).
- The equivalent uniform annual series *A* for an arithmetic gradient *G* is found by multiplying (*P/G*, *i*%, *n*) by (A/P,i%,n)

$$A = G(P/G, i\%, n)(A/P, i\%, n)$$

$$= G(A/G, i\%, n)$$

$$A = \frac{G}{i}\left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n}\right]\left[\frac{i(1+i)^n}{(1+i)^n - 1}\right]$$

$$= G\left[\frac{1}{i} - \frac{n}{(1+i)^n - 1}\right]$$

$$\left[\frac{1}{i} - \frac{n}{(1+i)^n - 1}\right]$$

- is the arithmetic-gradient uniform-series factor and is identified by (A/G, $i\%$, n).

$$A_G = G\left[\frac{1}{i} - \frac{n}{(1+i)^n - 1}\right]$$

- An *F/G* factor (arithmetic-gradient future worth factor)can be found by multiplying *P/G* and *F/P* factors. The resulting (*F/G*, *i*%, *n*) is:

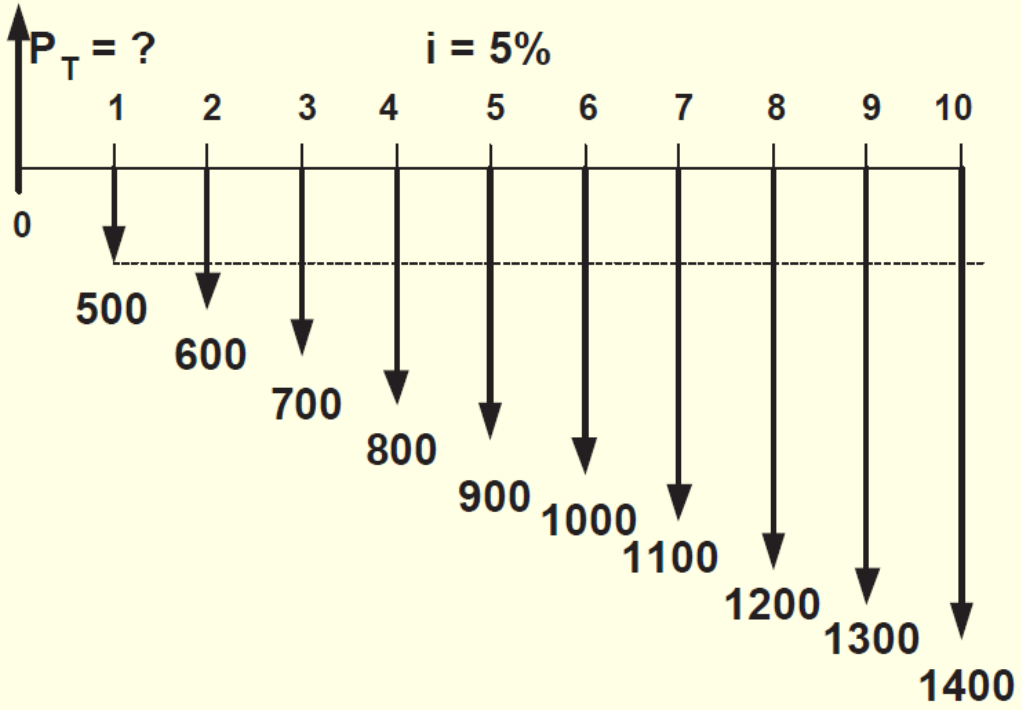
$$
\begin{aligned}
F &= G(P/G, i\%, n)(F/P, i\%, n) \\
&= G(F/G, i\%, n) \\
&= \frac{G}{i}\left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n}\right](1+i)^n \\
&= G\left[\frac{1}{i}\left(\frac{(1+i)^n - 1}{i} - n\right)\right]
\end{aligned}
$$

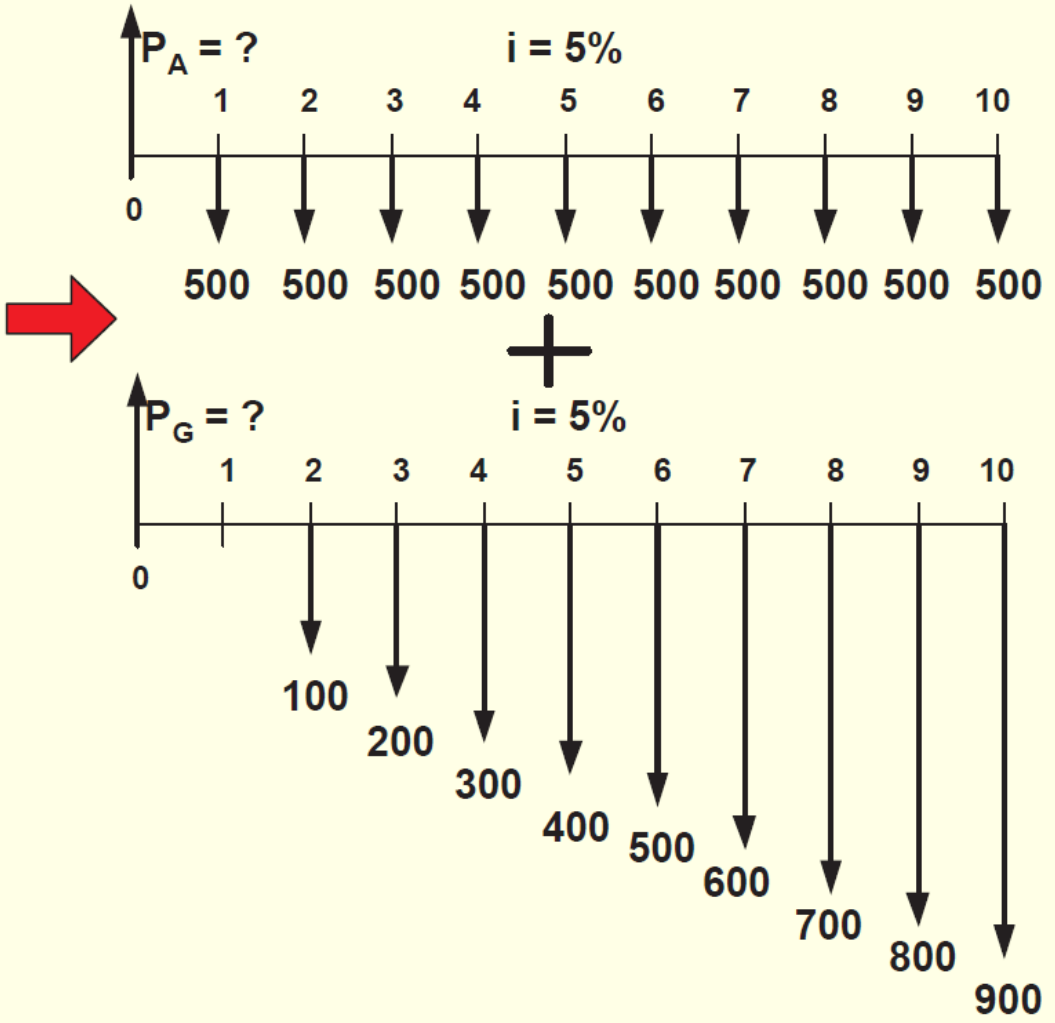
$$
F_G = G\left(\frac{1}{i}\right)\left[\left(\frac{(1+i)^n - 1}{i}\right) - n\right]
$$

- The total present worth P_T of a gradient series must consider the base and the gradient separately.
- Let P_A be the present worth of the base amount; uniform series amount A starting from the end of period 1 extending through period n.
- Let P_G be the present worth of an increasing gradient and $-P_G$ be the present worth of an decreasing gradient.
- Therefore: $P_T = P_A + P_G$ and $P_T = P_A - P_G$ for increasing and decreasing gradient series, respectively.

EXAMPLE

- Three counties in Florida agreed to pool tax resources already designated for county-maintained bridge refurbishment. At a recent meeting, the county engineers estimated that a total of $500,000 will be deposited at the end of next year into an account for the repair of old bridges throughout the three-county area.
- Further, they estimate the deposits will increase by $100,000 per year for only 9 years thereafter, then cease.
- Determine the equivalent (*a*) present worth, and (*b*) annual series amounts, if public funds earn at a rate of 5% per year.





$P_T = P_A + P_G$, we have

$G = 100$ and $A = 500$.

Solution by Standard Notation

$$\begin{aligned} P_T &= 500(P/A, 5\%, 10) + 100(P/G, 5\%, 10) \\ &= 500(7.7217) + 100(31.6520) = 7026,05 => \$7,026,050 \end{aligned}$$

Solution by Formulas

$$\begin{aligned} P_T &= 500\left[\frac{(1+i)^n - 1}{i(1+i)^n}\right] + 100\frac{1}{i}\left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n}\right] \\ &= 500\left[\frac{(1.05)^{10} - 1}{0.05(1.05)^{10}}\right] + 100\frac{1}{0.05}\left[\frac{(1.05)^{10} - 1}{0.05(1.05)^{10}} - \frac{10}{(1.05)^{10}}\right] \\ &= 500(7.7217) + 100(31.6520) = 7026,05 => \$7,026,050 \end{aligned}$$

- *(b)* consider the gradient and the base amount separately.
- The total annual series A_T (occuring in years 1 through 10.) is found by

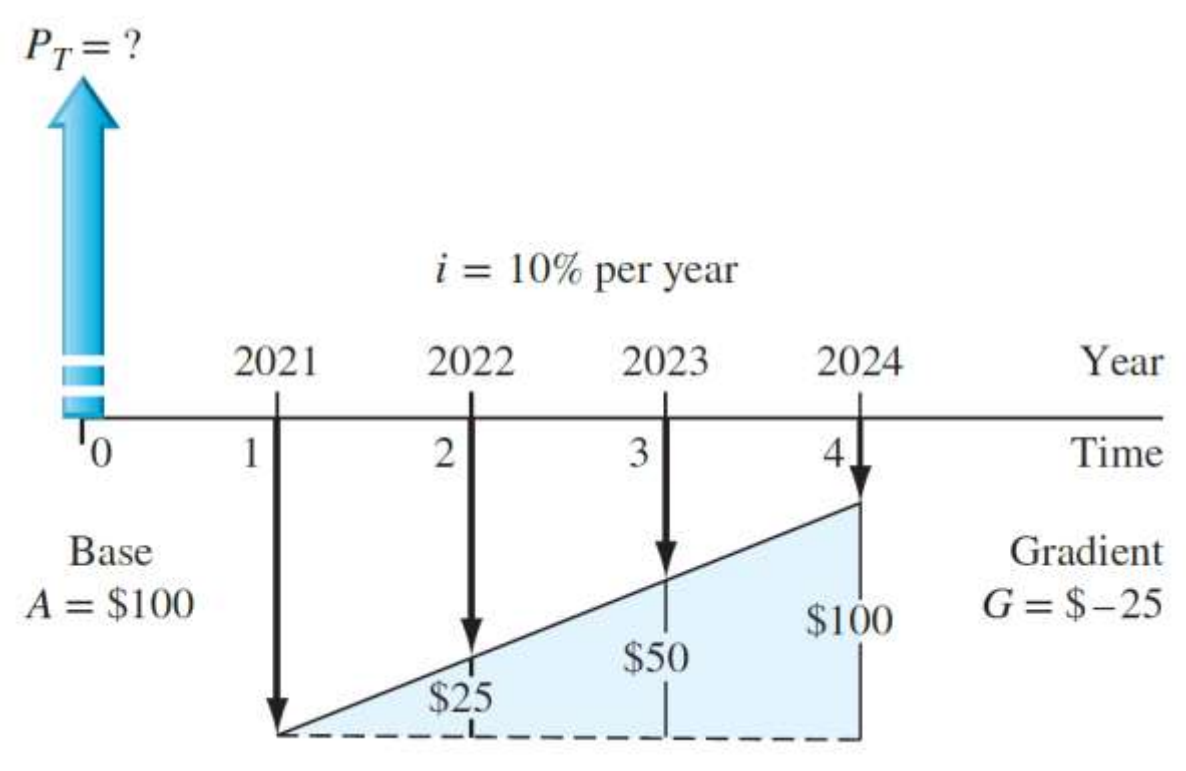
$$A_T = 500 + 100(A/G,5\%,10) = 500 + 100(4.0991)$$
$$= \$909.91 \text{ per year}$$

- **Comment**
- *Remember:* The *P/G* and *A/G* factors determine the present worth and annual series of the *gradient only.*
- Any other cash flows must be considered separately.
- If the present worth is already calculated [as in part *(a)*], *PT* can be multiplied by an *A/P* factor to get *AT*. In this case, considering round-off error,

$$A_T = P_T(A/P,5\%,10) = 7026.05(0.12950)$$
$$= \$909.873$$

EXAMPLE

- The announcement of the HBNA steel plant states that the $200 million (M) investment is planned for 2020. Most large investment commitments are actually spread out over several years as the plant is constructed and production is initiated. Further investigation may determine, for example, that the $200 M is a present worth in the year 2020 of anticipated investments during the next 4 years (2021 through 2024). Assume the amount planned for 2021 is $100 M with constant decreases of $25 M each year thereafter. Assume the time value of money for investment capital is 10% per year to answer the following questions
- (*a*) In equivalent present worth values, does the planned decreasing investment series equal the announced $200 M in 2020? Use both tabulated factors and spreadsheet functions.
- (*b*) Given the planned investment series, what is the equivalent annual amount that will be invested from 2021 to 2024?



- (*a*) The investment series is a decreasing arithmetic gradient with a base amount of \$100 M in year 1 (2021) and *G* = \$−25 M through year 4 (2024).
- Shaded area shows the constantly declining investment each year

$$P_T = P_A - P_G = 100(P/A,10\%,4) - 25(P/G,10\%,4)$$
$$= 100(3.1699) - 25(4.3781)$$
$$= \$207.537$$

- In present worth terms, the planned series will exceed the equivalent of \$200 M in 2020 by approximately \$7.5 M.

- (*b*) There are two equally correct ways to find A_T.

Use

$$A_T = A_A \pm A_G$$

$$A_T = 100 - 25(A/G,10\%,4) = 100 - 25(1.3812)$$

$$= \$65.471 \text{ per year}$$

Use P_T:

$$A_T = 207.537(A/P,10\%,4) = 207.537(0.31547)$$

$$= \$65.471 \text{ per year}$$

Geometric Gradient Series Factors

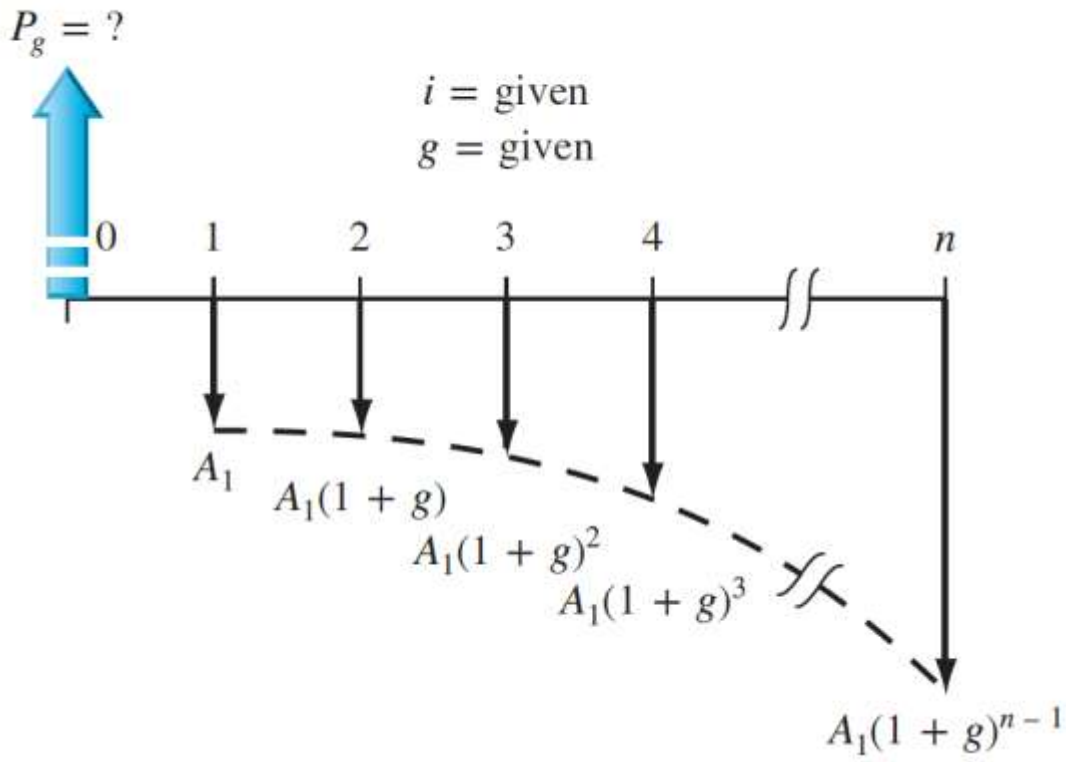
- Cash flow series that increase or decrease from period to period by constant percentage.
- g is the constant rate of change in decimal form, by which amounts change
- The series starts at the end of year 1, with amount $A1$, which is not considered as a base amount

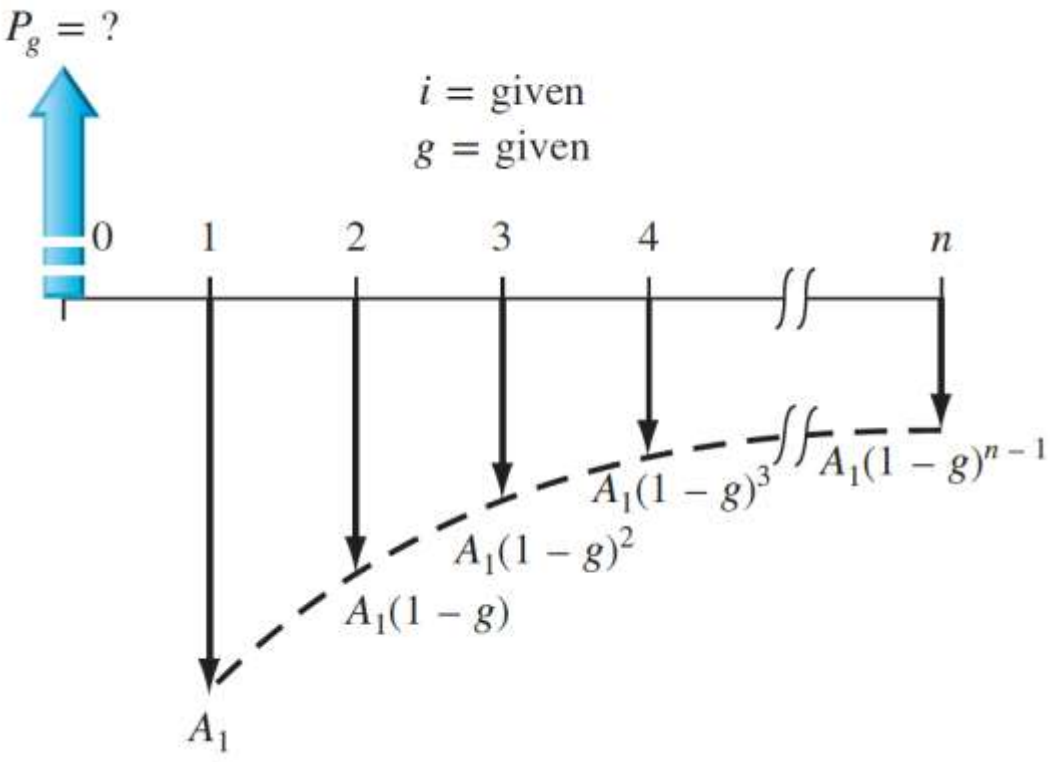
g = **constant rate of change,** in decimal form, by which cash flow values increase or decrease from one period to the next. The gradient g can be + or −.

A_1 = **initial cash flow in period 1** of the geometric series

P_g = **present worth** of the entire geometric gradient series, including the initial amount A_1

- Note that the initial cash flow A_1 **is not considered separately** when working with geometric gradients.
- Figure shows increasing and decreasing geometric gradients starting at an amount A_1 in time period 1 with present worth *Pg* located at time 0.





- The total present worth *Pg* for the entire cash flow series can be derived as:

$$
\begin{aligned}
P_g &= A_1\left[\frac{1}{(1+i)}\right] + A_1\left[\frac{(1+g)}{(1+i)^2}\right] + \cdots + A_1\left[\frac{(1+g)^{n-2}}{(1+i)^{n-1}}\right] \\
&+ A_1\left[\frac{(1+g)^{n-1}}{(1+i)^n}\right] \\
&= A_1\left[\frac{1}{(1+i)} + \frac{(1+g)}{(1+i)^2} + \cdots + \frac{(1+g)^{n-2}}{(1+i)^{n-1}} + \frac{(1+g)^{n-1}}{(1+i)^n}\right]
\end{aligned}
$$

$$P_g\left[\frac{(1+g)}{(1+i)}\right] = A_1\left[\frac{1+g}{(1+i)^2} + \frac{(1+g)^2}{(1+i)^3} + \cdots + \frac{(1+g)^{n-1}}{(1+i)^n} + \frac{(1+g)^n}{(1+i)^{n+1}}\right]$$

$$P_g\left[\frac{(1+g)}{(1+i)} - 1\right] = A_1\left[\frac{(1+g)^n}{(1+i)^{n+1}} - \frac{1}{(1+i)}\right]$$

$$P_g(g-i) = A_1\left[\frac{(1+g)^n}{(1+i)^n} - 1\right]$$

Therefore, when $i \neq g$;

$$P_g = A_1 \left[\frac{\left[1 - \frac{(1+g)^n}{(1+i)^n} \right]}{(i-g)} \right]$$

when $i = g$;

$$P_g = A_1 \left[\frac{1}{(1+i)} + \frac{1}{(1+i)} + \cdots + \frac{1}{(1+i)} + \frac{1}{(1+i)} \right]$$

$$= A_1 \left[\frac{n}{(1+i)} \right]$$

- These two factors, depending on the case whether $g = i$ or not, will transform the geometric-gradient series starting at the end of period 1 with amount $A1$ that is changing with constant rate g to the the present worth Pg at time 0. The standard notation for the factor is $(P/A, g, i, n)$.

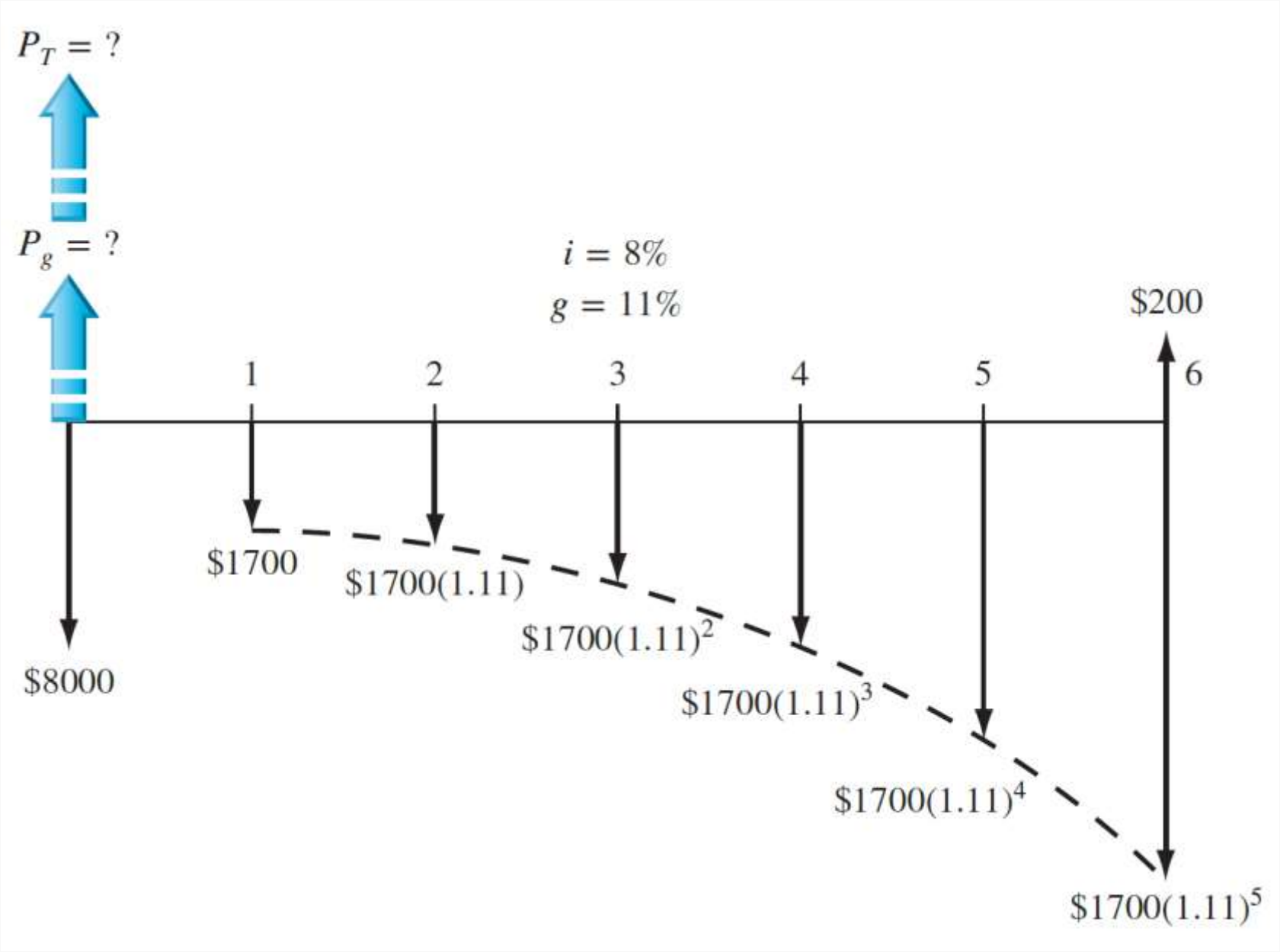
$$P_g = A_1(P/A,g,i,n)$$

$$(P/A,g,i,n) = \begin{cases} \dfrac{1-\left(\dfrac{1+g}{1+i}\right)^n}{i-g} & g \neq i \\ \dfrac{n}{1+i} & g = i \end{cases}$$

- These two factors, depending on the case whether $g = i$ or not, will transform the geometric-gradient series starting at the end of period 1 with amount $A1$ that is changing with constant rate g to the the present worth Pg at time 0. The standard notation for the factor is $(P/A, g, i, n)$.
- It is possible to derive factors for the equivalent A and F values; however, it is easier to determine the Pg amount and then multiply by the A/P or F/P factor.

EXAMPLE

- A coal-fired power plant has upgraded an emission control valve. The modification costs only $8000 and is expected to last 6 years with a $200 salvage value.
- The maintenance cost is expected to be high at $1700 the first year, increasing by 11% per year thereafter.
- Determine the equivalent present worth of the modification and maintenance cost at 8% per year.



- The cash flow diagram shows the salvage value as a positive cash flow and all costs as negative.

- For $g \neq i$ to calculate Pg. Total P_T is the sum of three present worth components.

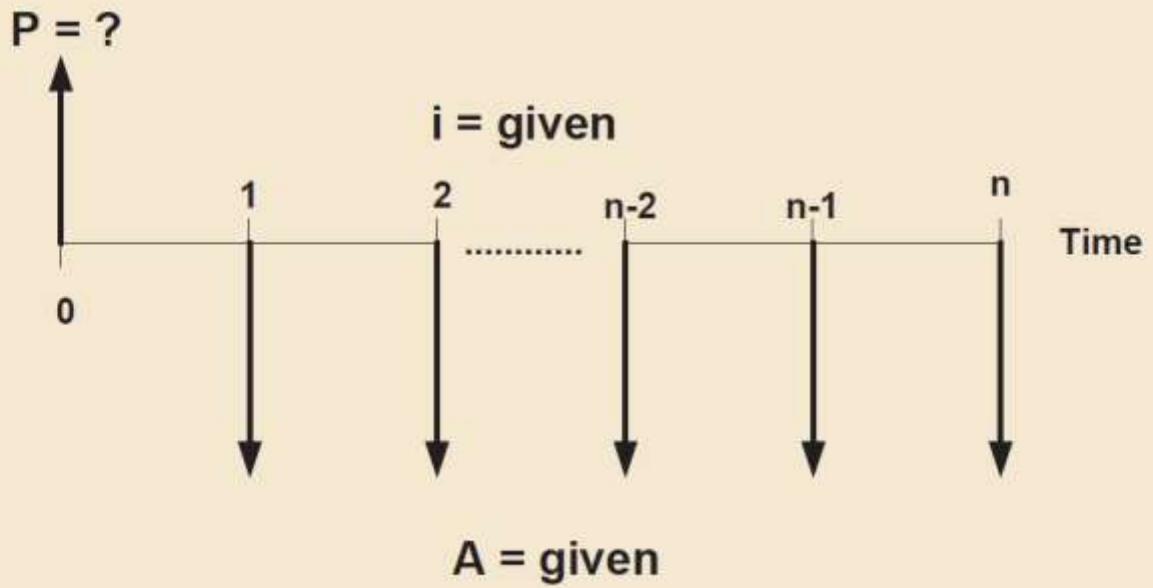
$$P_T = -8000 - P_g + 200(P/F,8\%,6)$$

$$= -8000 - 1700\left[\frac{1-(1.11/1.08)^6}{0.08-0.11}\right] + 200(P/F,8\%,6)$$

$$= -8000 - 1700(5.9559) + 126 = \$-17{,}999$$

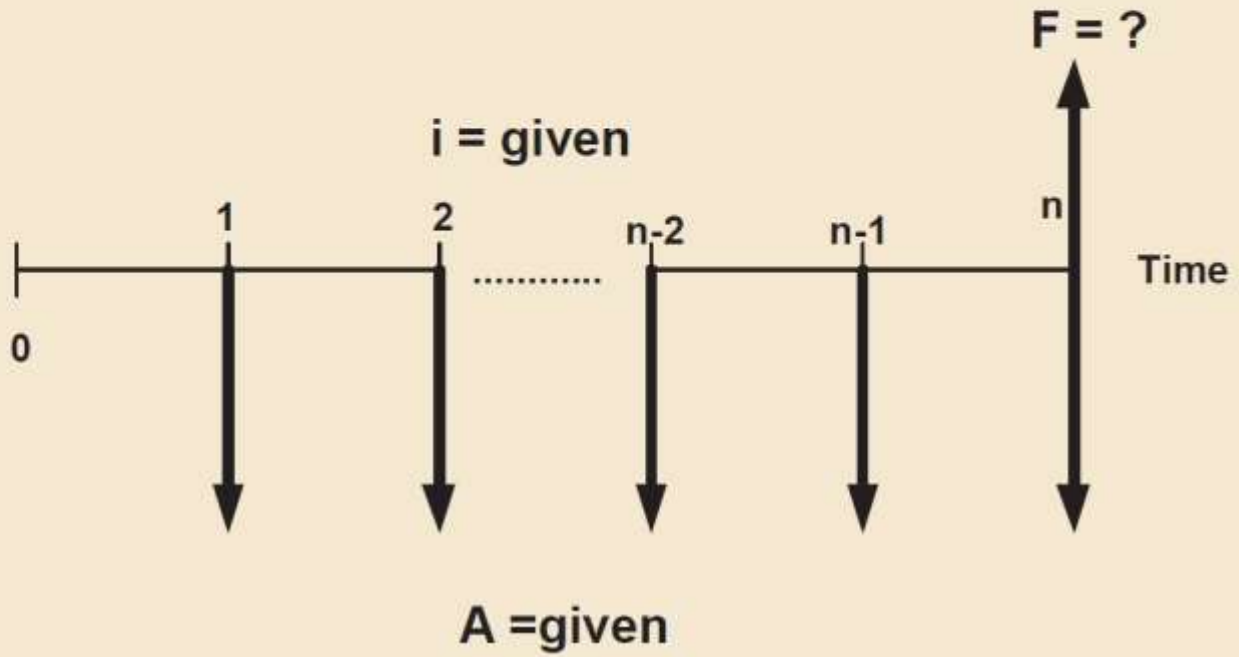
CAUTION on using P/A factor: $(P/A, i\%, n)$

The present worth P and the first uniform annual amount A are one period apart. Therefore, the present worth P must always be located one period prior to the first A.



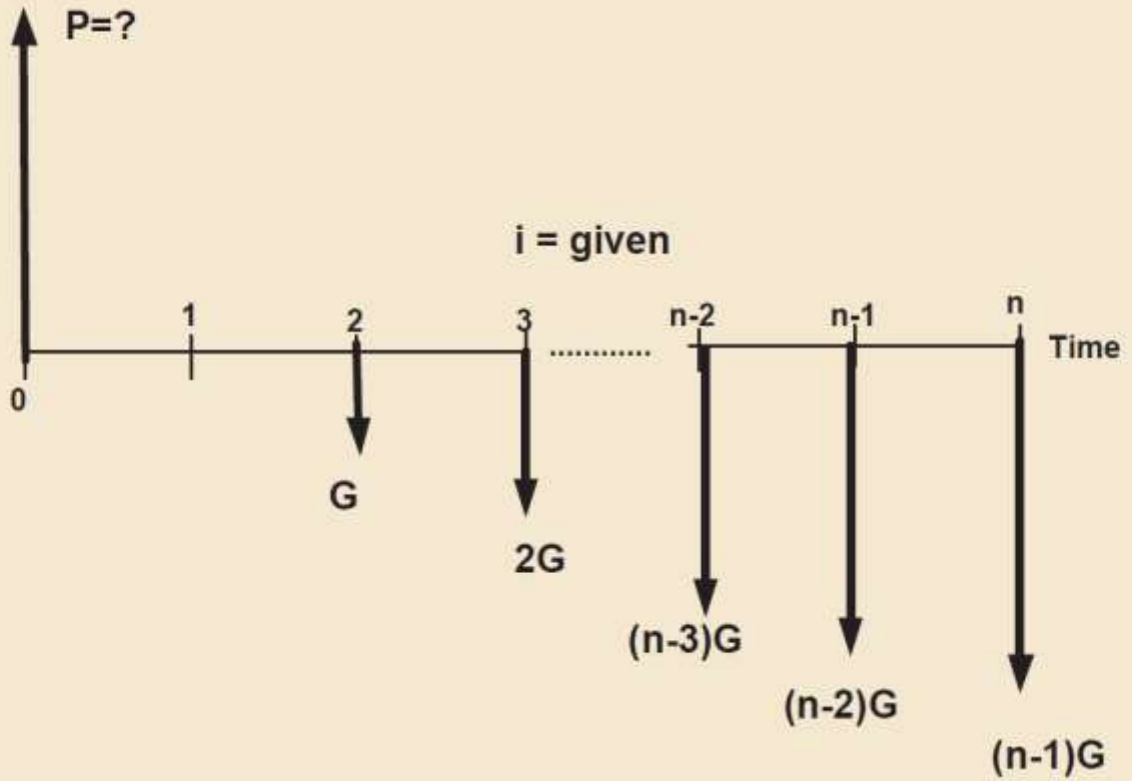
CAUTION on using F/A factor: $(F/A, i\%, n)$

The uniform series A begins at the end of period 1 and continues through the period n. Therefore, the future worth F is at the same period as the last A.



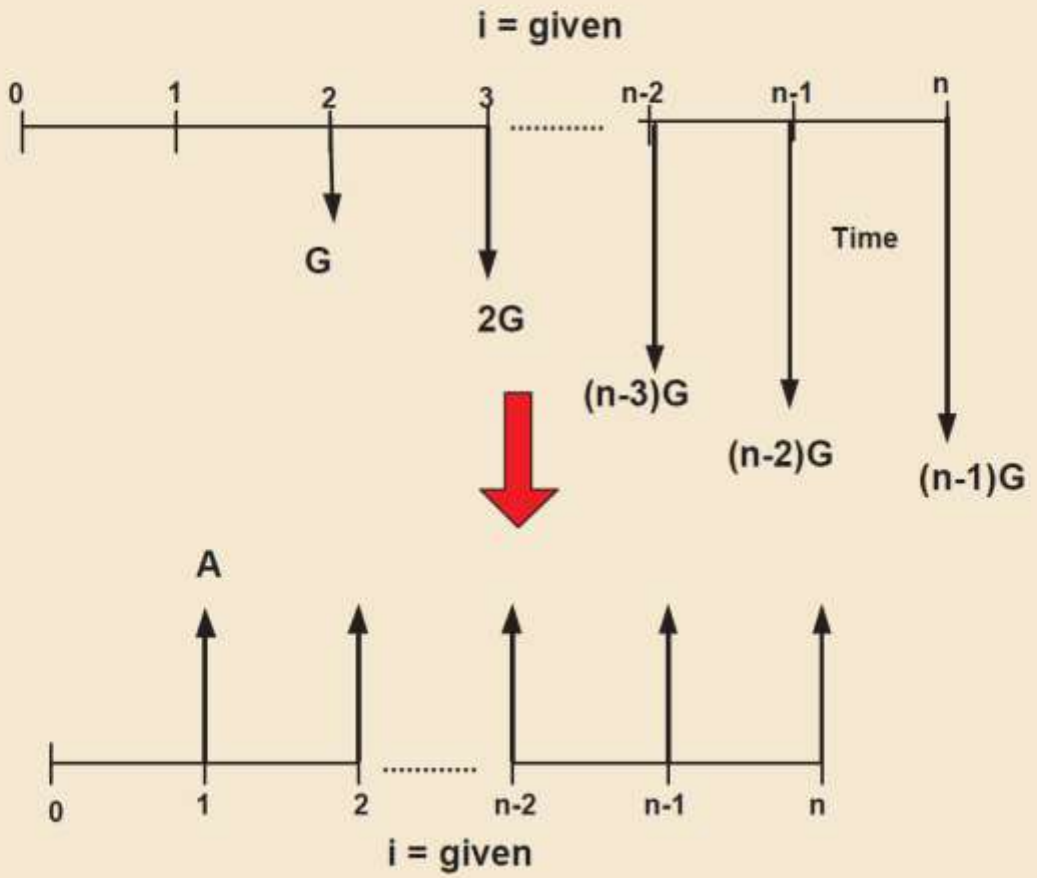
CAUTION on using P/G factor: : $(P/G, i\%, n)$

The gradient begins between years 1 and 2, continues through the period *n*. Therefore, the present worth of an arithmetic gradient will always be located two periods before the gradient starts.



CAUTION on using A/G factor: : $(A/G, i\%, n)$

The gradient begins between years 1 and 2, continues through the period *n*. Therefore, the equivalent annual series of an arithmetic gradient will start from period 1 and continue through the period *n*.

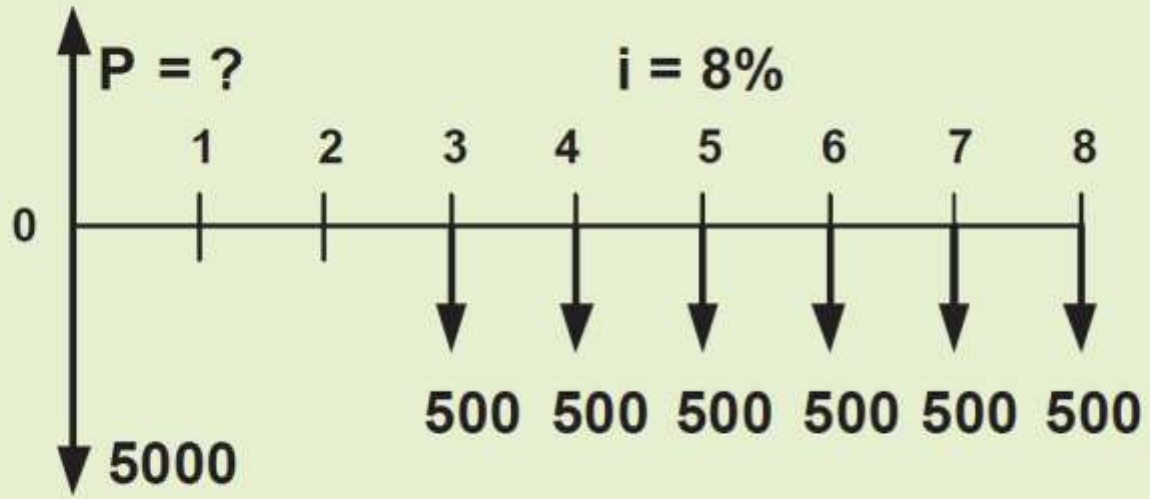


SHIFTED SERIES

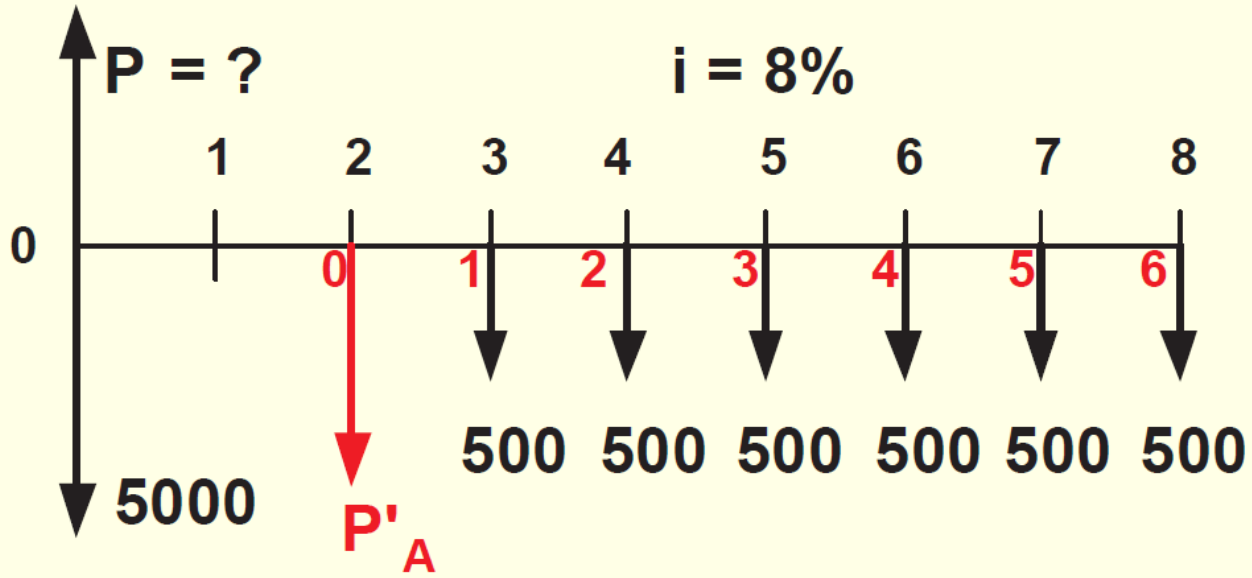
- Most estimated cash flow series do not fit exactly the series for which the factors are derived so far.
- Therefore, we need to combine or shift the given series.
- Renumbering the cash flow diagram is a good practice.

Example

The engineering company just purchased new CAD software for $5,000 now and annual payments of $500 per year for 6 years, starting 3 years from now for annual upgrades. What is the present worth of the payments if the interest rate is 8% per year.



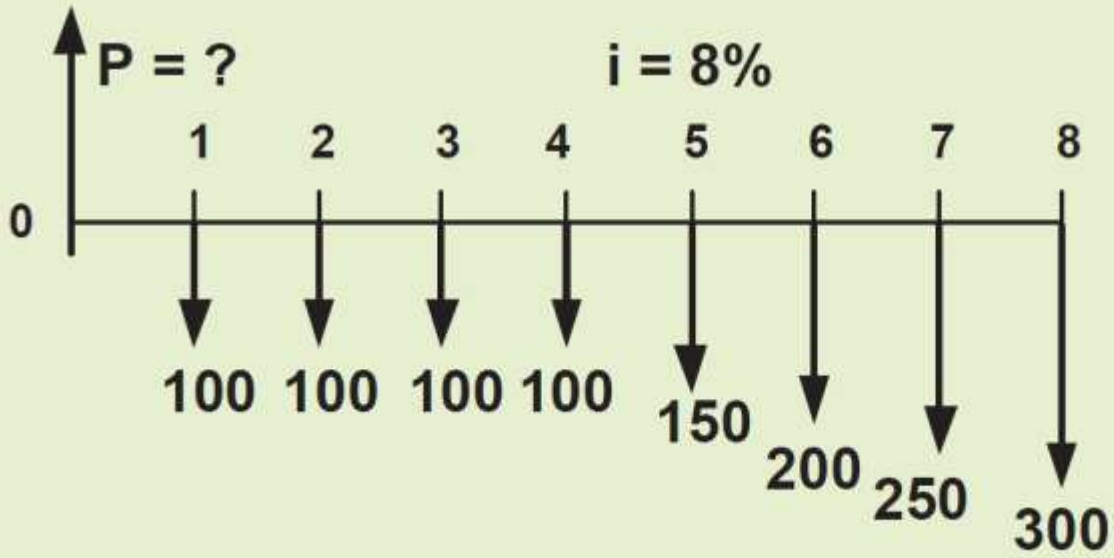
$$P = 5000 + P_A$$

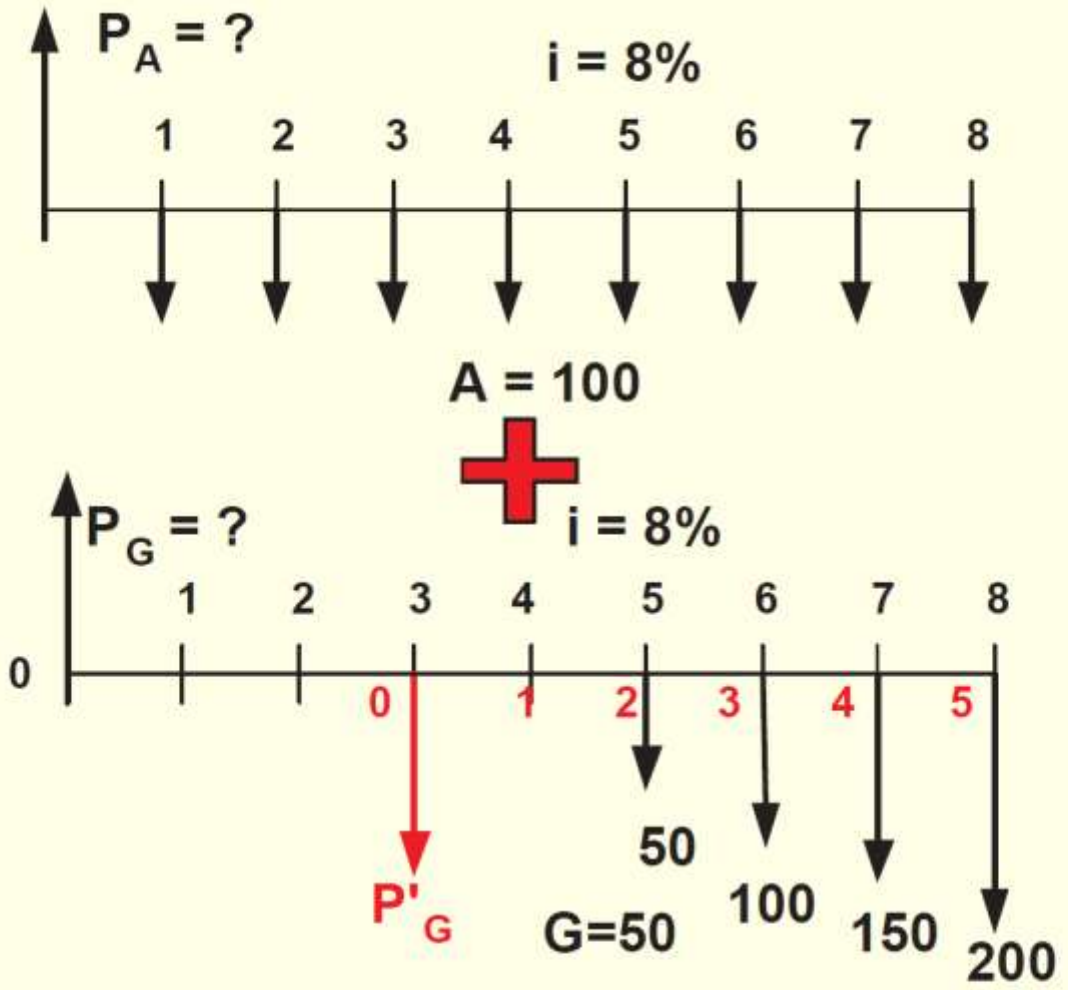


$$
\begin{aligned}
P &= 5000 + P'_A(P/F, 8\%, 2) \\
&= 5000 + 500(P/A, 8\%, 6)(P/F, 8\%, 2) \\
&= 5000 + 500(4.6229)(0.8573) \\
&= \$6981.60
\end{aligned}
$$

Example

The average inspection cost on robotics manufacturing line has been tracked for 8 years. Cost averages were steady at $ 100 per completed unit for the first 4 years, but have increased consistently by $50 per unit for each of the last 4 years. Find the present worth equivalent of the cost averages assuming that the applicable interest rate is 8% ?

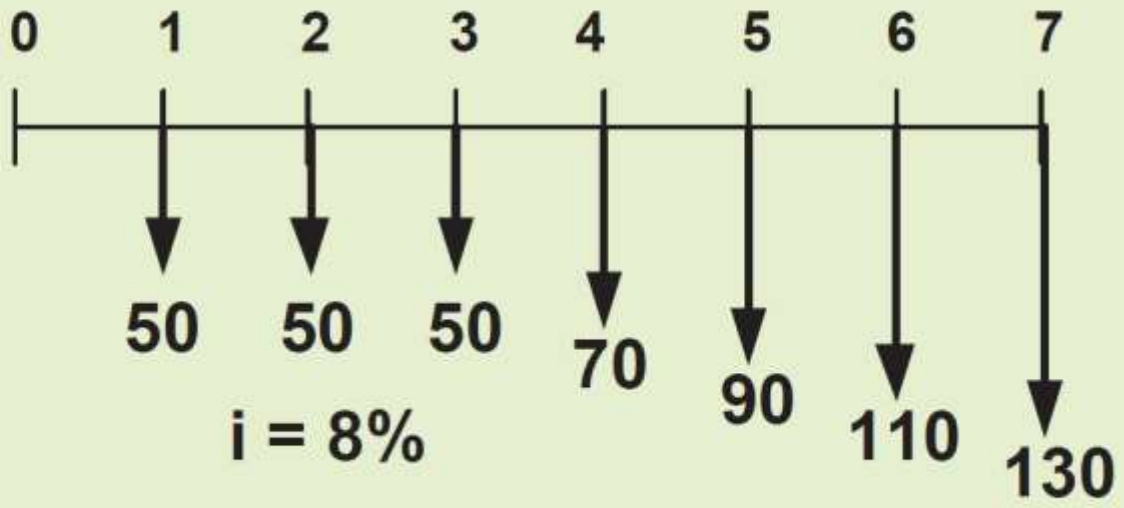


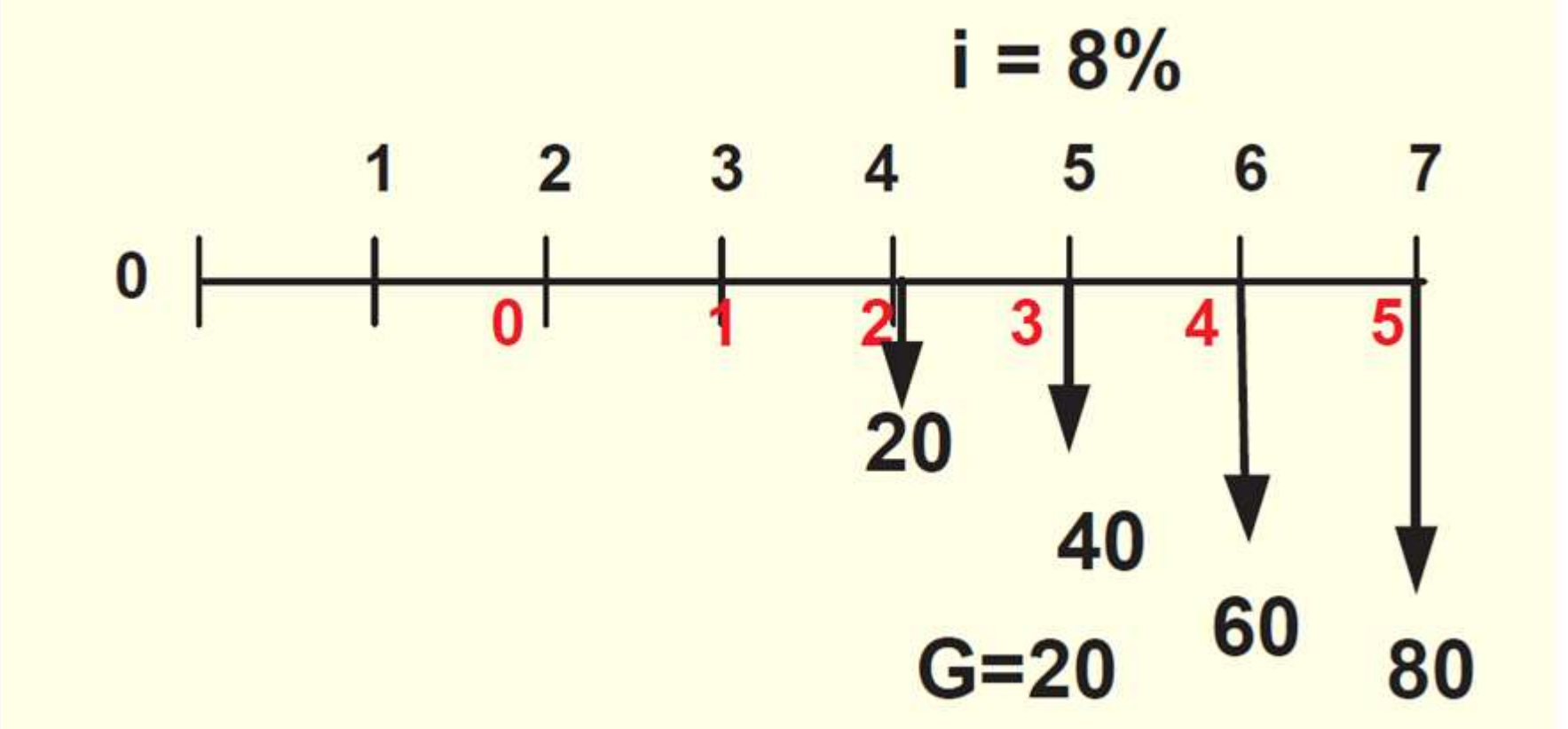


$$
\begin{aligned}
P &= P_A + P_G = P_A + P'_G(P/F, 8\%, 3) \\
&= 100(P/A, 8\%, 8) + 50(P/G, 8\%, 5)(P/F, 8\%, 3) \\
&= 100(5.7466) + 50(7.3724)(0.7938) = \$867.27
\end{aligned}
$$

Example

Compute the equivalent annual series in years 1 through 7 for the cash flow estimates given below when applicable interest rate is 8%.




i = 8%
1
2
3
4
5
6
7
0
0
1
2
3
4
5
20
40
60
80
G=20

The **present worth factor for uniform series** is:

$$P = A \cdot \frac{(1+i)^n - 1}{i(1+i)^n}$$

- $A = 50$
- $n = 3$
- $i = 0.08$

$$P_1 = 50 \cdot \frac{(1.08)^3 - 1}{0.08(1.08)^3} = 50 \cdot \frac{1.2597 - 1}{0.08 \cdot 1.2597} = 50 \cdot \frac{0.2597}{0.10078} \approx 50 \cdot 2.578 \approx 128.9$$

PV of gradient (years 4–7) at start of year 3):

$$P_G = G \cdot \frac{(1+i)^n - in - 1}{i^2(1+i)^n}$$

$$(1.08)^4 = 1.3605$$

$$(1.3605 - 0.08 \cdot 4 - 1)/(0.08^2 \cdot 1.3605) = (1.3605 - 0.32 - 1)/(0.0064 \cdot 1.3605) = 0.0405/0.008707 \approx 4.65$$

$$P_G = 20 \cdot 4.65 \approx 93.0$$

This PV is **at the end of year 3**, so we must **discount it back 3 years to year 0**:

$$PV_{G0} = \frac{93.0}{(1.08)^3} = \frac{93.0}{1.2597} \approx 73.8$$

Total Present Value

$$PV_{\text{total}} = PV_{1-3} + PV_{G0} = 128.9 + 73.8 \approx 202.7$$

Convert to Equivalent Annual Series (7 years)

Uniform annual series formula:

$$A = PV \cdot \frac{i(1+i)^n}{(1+i)^n - 1}$$

- $n = 7$ years, $i = 0.08$, $PV = 202.7$

$$A = 202.7 \cdot \frac{0.08 \cdot 1.08^7}{1.08^7 - 1} = 202.7 \cdot \frac{0.08 \cdot 1.7137}{1.7137 - 1} = 202.7 \cdot \frac{0.1371}{0.7137} \approx 202.7 \cdot 0.1922 \approx 38.95$$